Fluid Mechanics

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States of Matter

- Solid

  has a definite volume and shape

- Liquid

  has a definite volume but not a definite shape

- Gas – unconfined

  has neither a definite volume nor shape

Fluids

- is a collection of molecules that are randomly arranged and held together by weak cohesive forces and by forces exerted by the walls of a container.
- *Both liquids and gases are fluids*
- Fluids do not sustain shearing stresses or tensile stresses.
- The force exerted by a static fluid on an object is always perpendicular to the surfaces of the object.

Fluid Statics and Dynamics

- Fluid Statics
  - Describes fluids at rest
- Fluid Dynamics
  - Describes fluids in motion

Density

- Density is defined as the mass per unit volume of the substance.
- The values of density for a substance vary slightly with temperature since volume is temperature dependent.
- The various densities indicate the average molecular spacing in a gas is much greater than that in a solid or liquid.

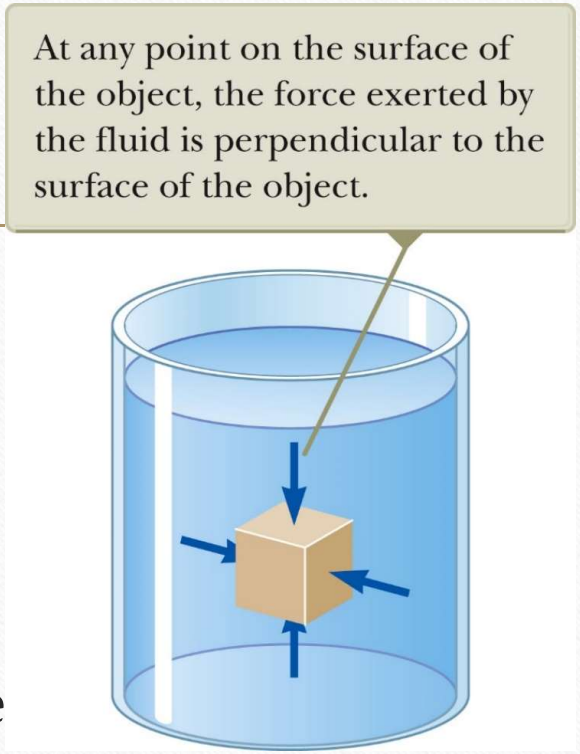
$$\rho = \frac{m}{V}.$$

TABLE 14.1 *Densities of Some Common Substances at Standard Temperature (0°C) and Pressure (Atmospheric)*

Substance	ρ (kg/m^3)	Substance	ρ (kg/m^3)
Air	1.29	Iron	7.86×10^3
Air (at 20°C and atmospheric pressure)	1.20	Lead	11.3×10^3
Aluminum	2.70×10^3	Mercury	13.6×10^3
Benzene	0.879×10^3	Nitrogen gas	1.25
Brass	8.4×10^3	Oak	0.710×10^3
Copper	8.92×10^3	Osmium	22.6×10^3
Ethyl alcohol	0.806×10^3	Oxygen gas	1.43
Fresh water	1.00×10^3	Pine	0.373×10^3
Glycerin	1.26×10^3	Platinum	21.4×10^3
Gold	19.3×10^3	Seawater	1.03×10^3
Helium gas	1.79×10^{-1}	Silver	10.5×10^3
Hydrogen gas	8.99×10^{-2}	Tin	7.30×10^3
Ice	0.917×10^3	Uranium	19.1×10^3

Pressure

- The **pressure** P of the fluid at the level to which the device has been submerged is the ratio of the force to the area..
- Unit of pressure is **pascal** (Pa)= N/m^2
- Pressure is a scalar and force is a vector.
- The direction of the force producing a pressure is perpendicular to the area of interest.



$$\text{pressure} = P = \frac{F}{A}.$$

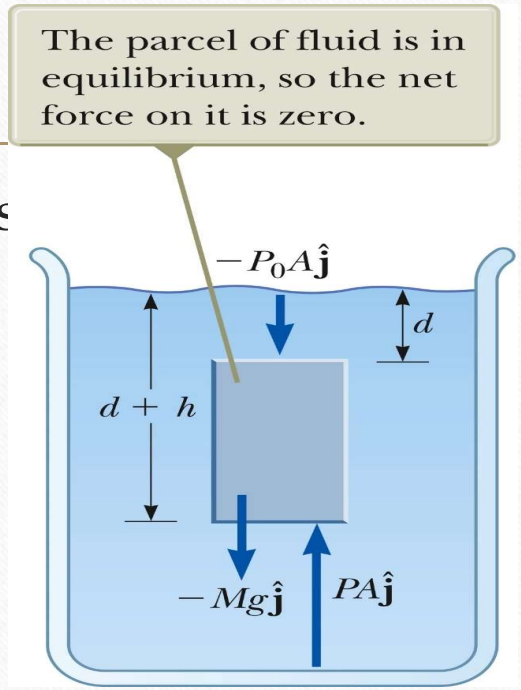
Variation of Pressure with Depth

- Fluids have pressure that varies with depth.
- If a fluid is at rest in a container, all portions of the fluid must be in static equilibrium.
- All points at the same depth must be at the same pressure.
  - Otherwise, the fluid would not be in equilibrium.
    - This is independent of the shape of the container.

Pressure and Depth

- Examine the darker region, a sample of an incompress liquid within a cylinder and has a density ρ
- The three forces are:
  - Downward force on the top, P_0A
  - Upward on the bottom, PA
  - Gravity acting downward, Mg
    - The mass can be found from the density:

$$M = \rho V = \rho A h.$$

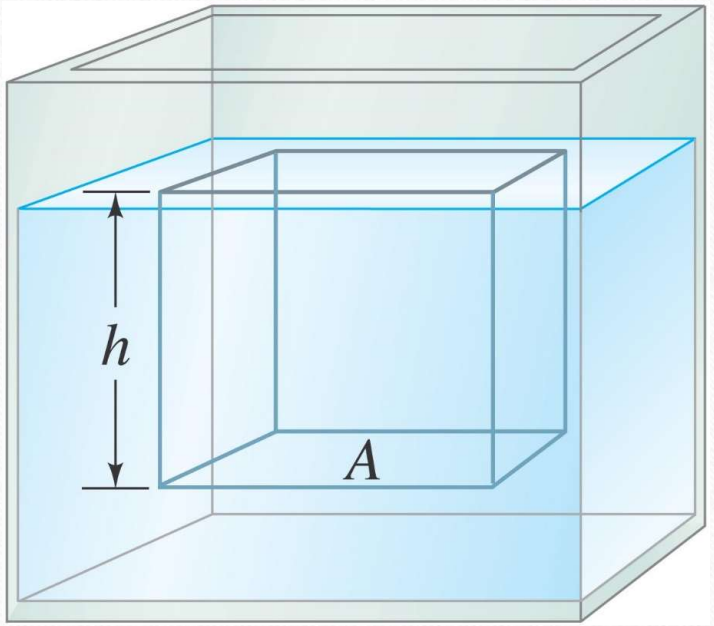


Pressure and Depth

The pressure at a depth h below the surface of the liquid is due to the weight of the liquid above it. We can quickly calculate:

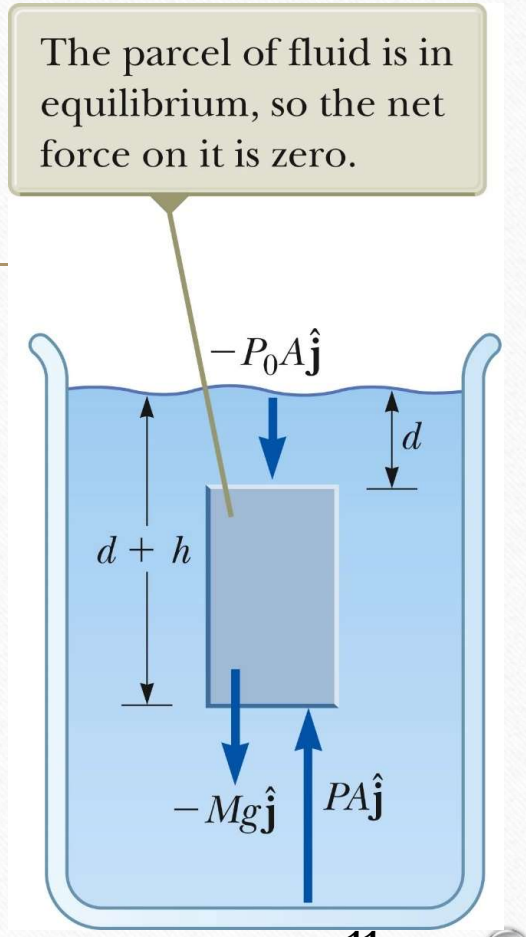
$$P = \frac{F}{A} = \frac{\rho A h g}{A}$$

$$P = \rho g h.$$



Pressure and Depth

- The absolute pressure at depth h is
  - $P = P_0 + \rho g h$
- The pressure P at a depth h below a point in the liquid at which the pressure is P_0 is greater by an amount $\rho g h$.



Atmospheric Pressure

- If the liquid is open to the atmosphere, and P_0 is the pressure at the surface of the liquid, then P_0 is *atmospheric pressure*.
- $P_0 = 1.00 \text{ atm} = 1.013 \times 10^5 \text{ Pa}$

Absolute vs. Gauge Pressure

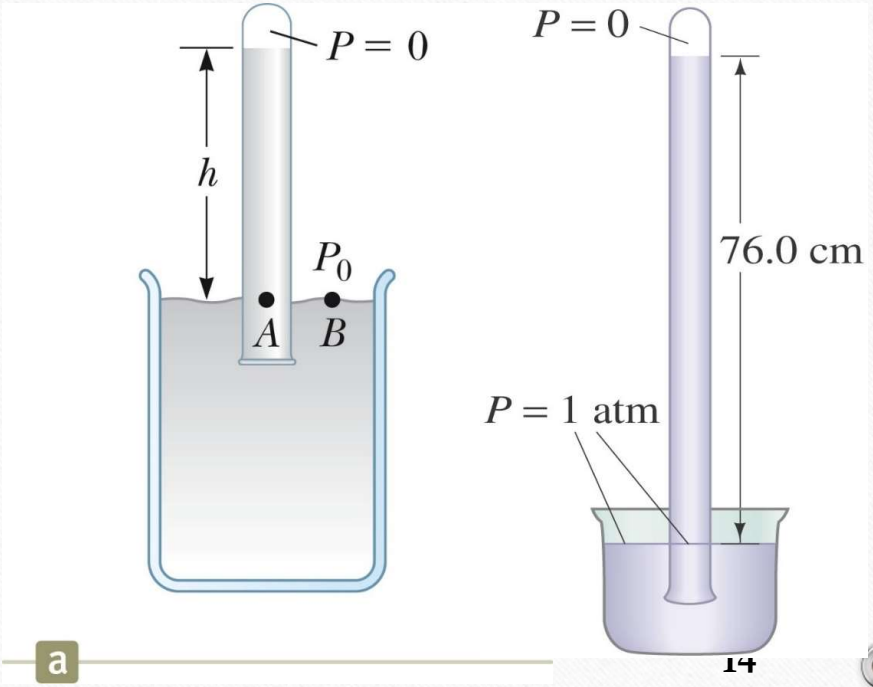
- $P = P_0 + \rho g h$
- P is the **absolute pressure.**
- The **gauge pressure** is $P - P_0$.
  - This is also $\rho g h$.
  - This is what you measure in your tires.

Pressure Measurements: Barometer

- A long closed tube is filled with mercury and inverted in a dish of mercury.
  - The closed end is nearly a vacuum.
- Measures atmospheric pressure as

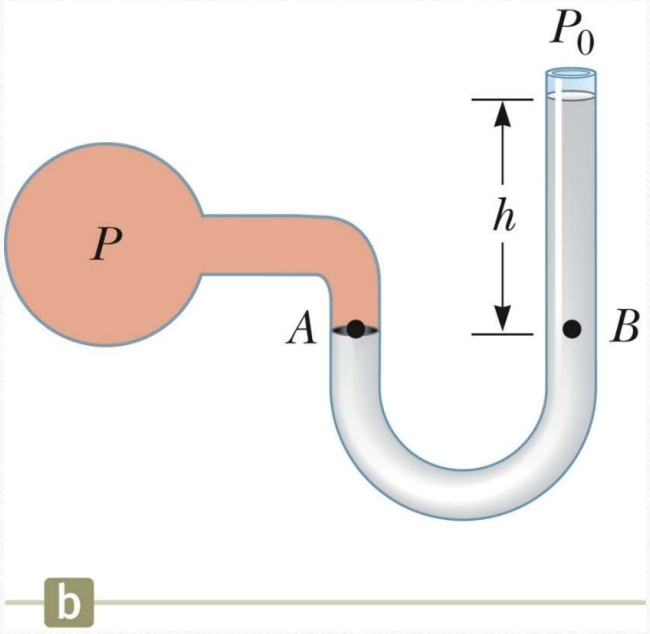
$$P_o = \rho_{Hg} g h$$

- One 1 atm = 0.760 m (of Hg)



Pressure Measurements: Manometer

- A device for measuring the pressure of a gas contained in a vessel.
- One end of the U-shaped tube is open to the atmosphere.
- The other end is connected to the pressure to be measured.
- Pressure at B is $P = P_o + \rho g h$
- The height can be calibrated to measure the pressure.



Pascal's Law

- The pressure in a fluid depends on depth and on the value of P_0.
- *An increase in pressure at the surface must be transmitted to every other point in the fluid.*
- This is the basis of Pascal's law.

$$P_1 = P_2 \qquad \frac{F_1}{A_1} = \frac{F_2}{A_2}$$

- **Pascal's Law:** *A change in the pressure applied to a fluid is transmitted undiminished to every point of the fluid and to the walls of the container.*

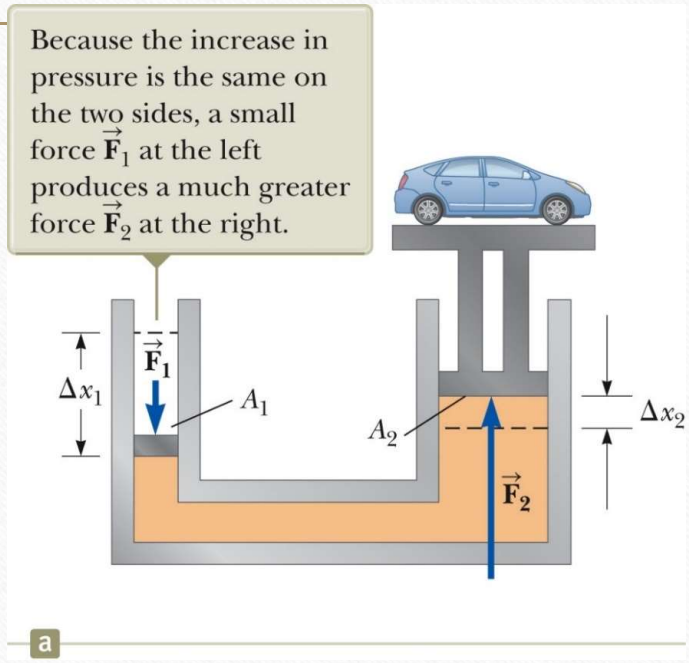
Pascal's Law

- A large output force can be applied by means of a small input force.

$$P_1 = P_2 \qquad \frac{F_1}{A_1} = \frac{F_2}{A_2}$$

- The volume of liquid pushed down on the left must equal the volume pushed up on the right.
- Since the volumes are equal,

$$A_1 \Delta x_1 = A_2 \Delta x_2$$

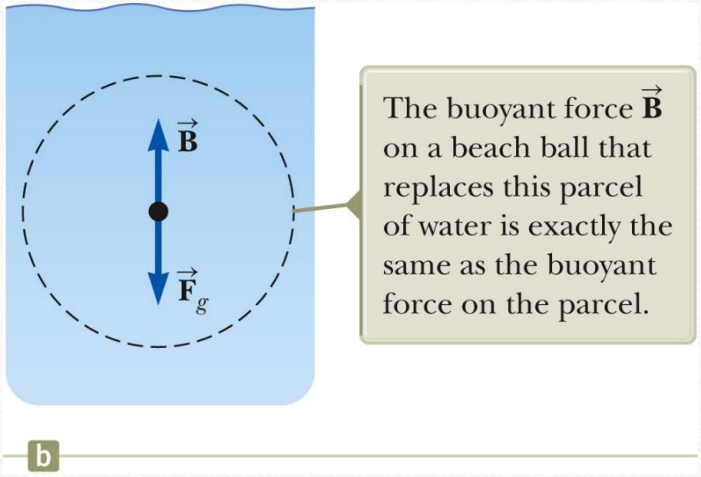


Pascal's Law

- Combining the equations,
  - $F_1 \Delta x_1 = F_2 \Delta x_2$ which means $Work_1 = Work_2$
  - This is a consequence of Conservation of Energy.
- If an external pressure is applied to a confined fluid, the pressure at every point within the fluid increases by that amount.
- This principle is also used in Hydraulic brakes, Car lifts, Hydraulic jacks, Forklifts

Buoyant Force

- The **buoyant force** is the upward force exerted by a fluid on any immersed object.
- The parcel is in equilibrium.
- There must be an upward force to balance the downward gravitational force.
- The magnitude of the upward (buoyant) force must equal (in magnitude) the downward gravitational force.
- The buoyant force is the resultant force due to all forces applied by the fluid surrounding the parcel.



Archimedes's Principle

- The magnitude of the buoyant force always equals *the weight of the fluid displaced by the object.*
  - This is called **Archimedes' Principle.**
- Archimedes' Principle does not refer to the makeup of the object experiencing the buoyant force.
  - The object's composition is not a factor since the buoyant force is exerted by the surrounding fluid.

Archimedes' Principle

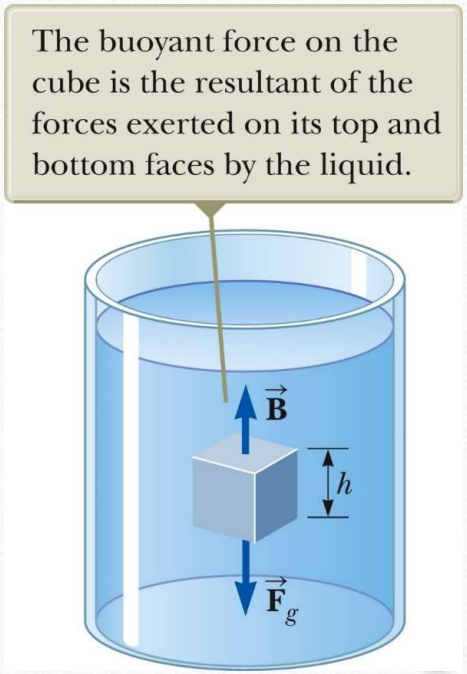
- The pressure at the bottom of the cube is greater than the pressure at the top of the cube.
- The pressure at the top of the cube causes a downward force of $P_{top}\,A$ and the pressure at the bottom of the cube causes an upward force of $P_{bot}\,A$.

$$F_B = (P_{bot} - P_{top})\,A = (\rho_{fluid}\,g\,h)\,A$$

$$F_B = \rho_{fluid}\;g\;V_{disp},$$

as $V_{disp} = A\,h$ is the volume of the fluid displaced by the cube.

$F_B = M\,g$, Mg is the weight of the fluid displaced by the cube.



Archimedes' Principle: Totally Submerged Object

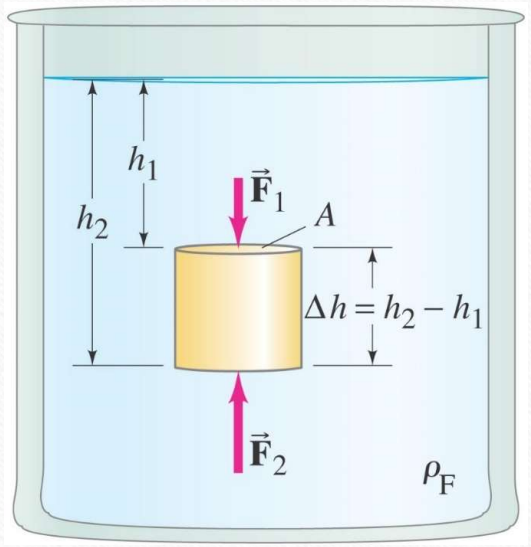
- An object is totally submerged in a fluid of density ρ_{fluid}.
- The volume V_{disp} of the fluid is equal to the volume of the object, V_{obj}.
- The upward buoyant force is

$$F_B = \rho_{fluid}\, g\, V_{object}$$

- The downward gravitational force is

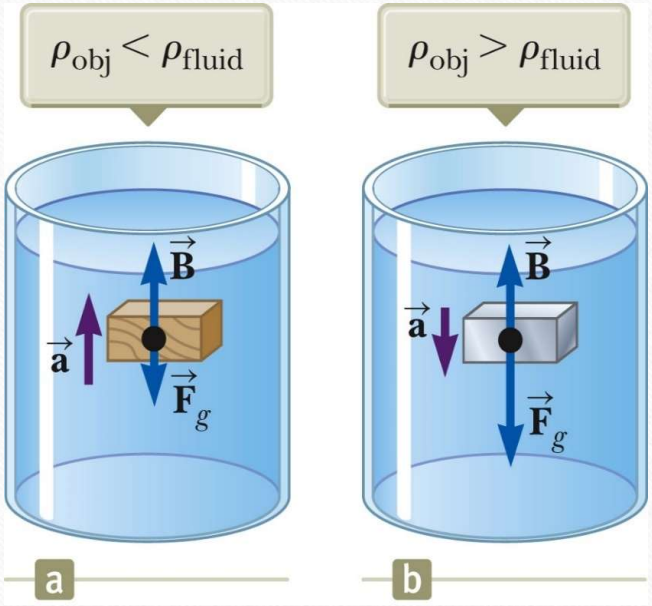
$$F_g = Mg = = \rho_{obj}\, g\, V_{obj}$$

- The net force is $F_B - F_g = (\rho_{fluid} - \rho_{obj})\, g\, V_{obj}$



Archimedes's Principle: Totally Submerged Object

- If the density of the object is less than the density of the fluid, the unsupported object accelerates upward.
- If the density of the object is more than the density of the fluid, the unsupported object sinks.
- If the density of the submerged object equals the density of the fluid, the object remains in equilibrium.
- The direction of the motion of an object in a fluid is determined only by the densities of the fluid and the object.



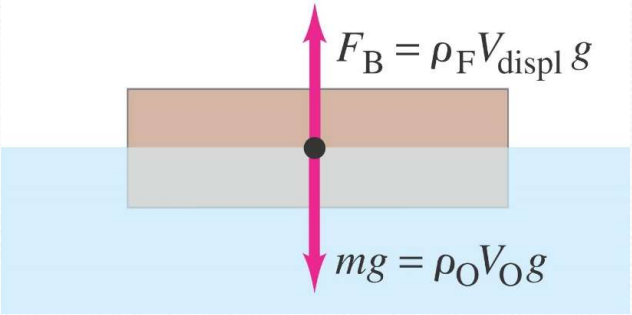
Archimedes's Principle: Floating Object

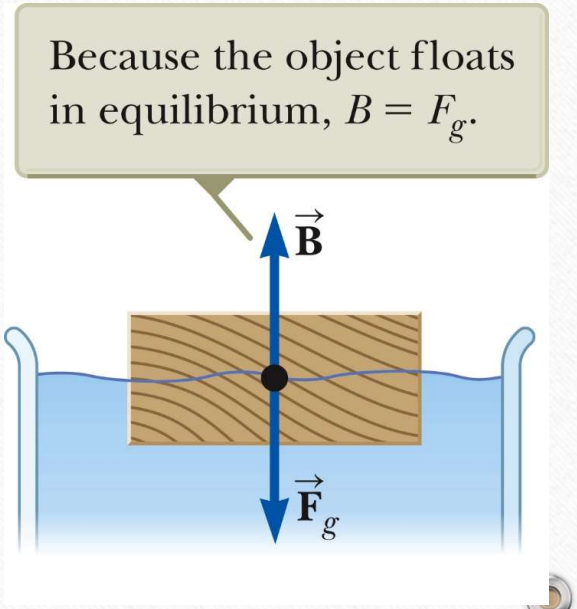
- The density of the object is less than the density of the fluid.
- The object is in static equilibrium.
- The object is only partially submerged.
- The upward buoyant force is balanced by the downward force of gravity.
- Volume of the fluid displaced corresponds to the volume of the object beneath the fluid level.

Archimedes' Principle: Floating Object

The fraction of the volume of a floating object that is below the fluid surface is equal to the ratio of the density of the object to that of the fluid.

$$\frac{V_{disp}}{V_{obj}} = \frac{\rho_{obj}}{\rho_{fluid}}$$





Example 1:

- **Car Lift**

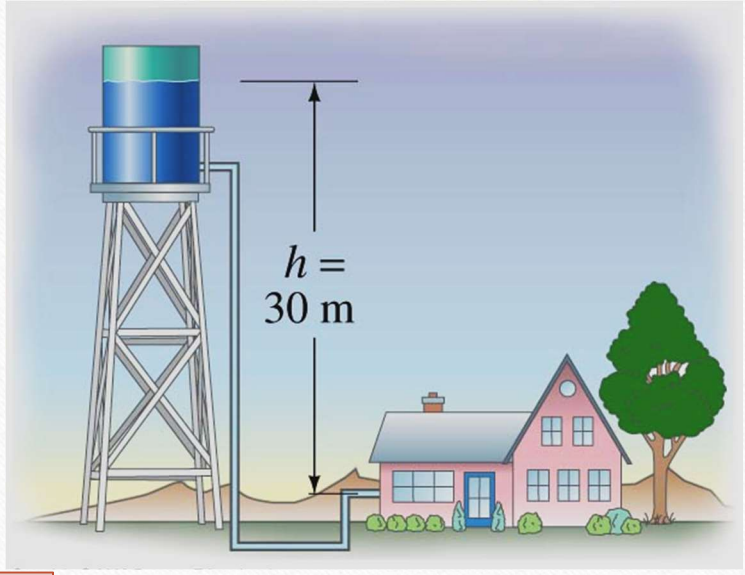
In a car lift used in a service station, compressed air exerts a force on a small piston that has a circular section of radius 5 cm. this pressure is transmitted by a liquid to a piston of radius 15 cm. what force must the compressed air exert to lift a car weighing 13300N? And the pressure that produces this force

1.48×10^3 N & 1.88×10^5 N/m^2

Example 2:

- Pressure at a faucet.

The surface of the water in a storage tank is 30 m above a water faucet in the kitchen of a house. Calculate the difference in water pressure between the faucet and the surface of the water in the tank.

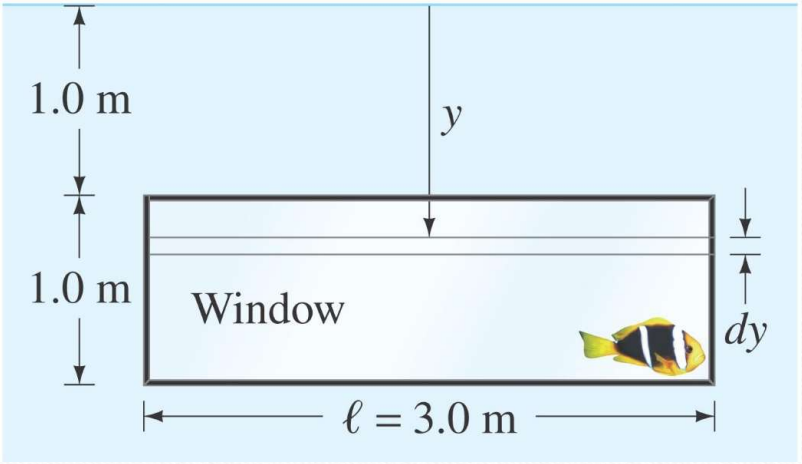


The pressure difference is 2.9×10^5 N/m^2

Example 3:

- Force on aquarium window.

Calculate the force due to water pressure exerted on a 1.0 m x 3.0 m aquarium viewing window whose top edge is 1.0 m below the water surface.

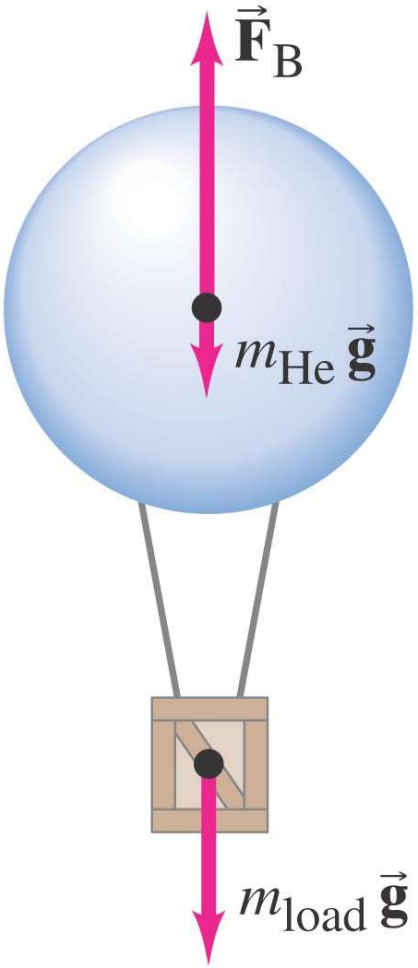


Example 4:

- Helium balloon.

What volume V of helium is needed if a balloon is to lift a load of 180 kg (including the weight of the empty balloon)?

160 m³.

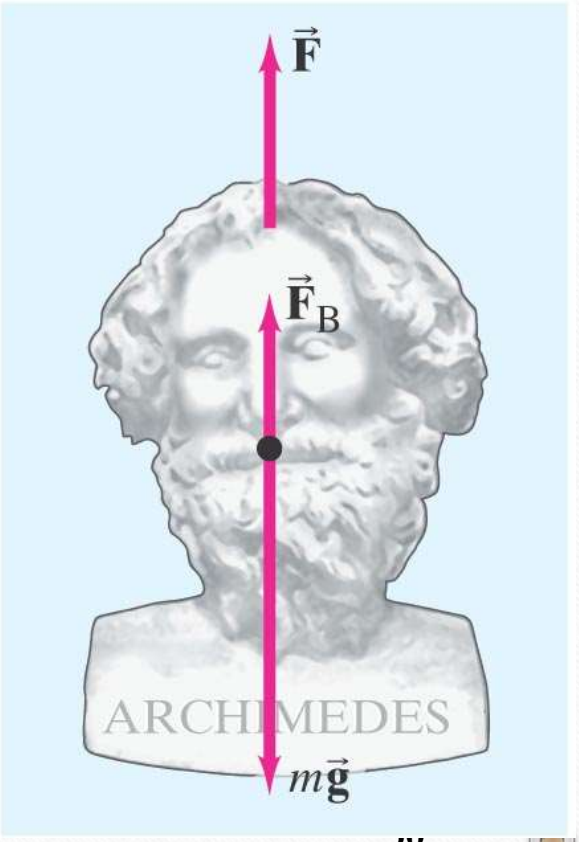


Example 5:

- **Recovering a submerged statue.**

A 70-kg ancient statue lies at the bottom of the sea. Its volume is 3.0×10^4 cm^3. How much force is needed to lift it?

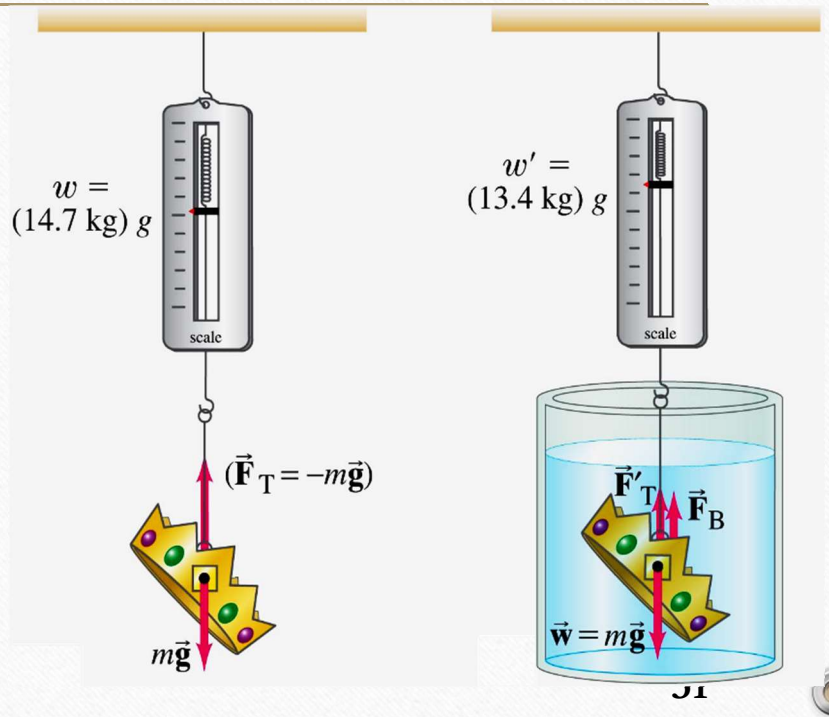
Force needed to lift the statue is 390 N



Example 6:

- Archimedes: Is the crown gold?

When a crown of mass 14.7 kg is submerged in water, an accurate scale reads only 13.4 kg. Is the crown made of gold?

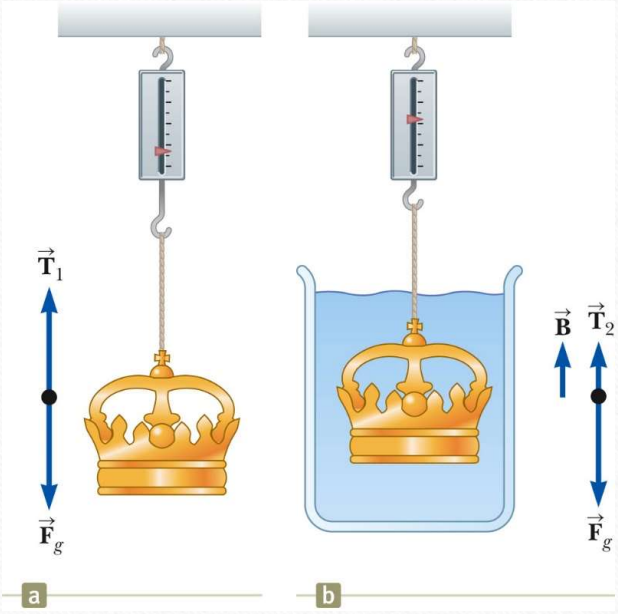


Example 5: Cont.

- Archimedes was (supposedly) asked, "Is the crown made of pure gold?"
- Crown's weight in air = 7.84 N
- Weight in water (submerged) = 6.84 N
- Buoyant force will equal the apparent weight loss
  - Difference in scale readings will be the buoyant force
- Categorize the crown as a particle in equilibrium.

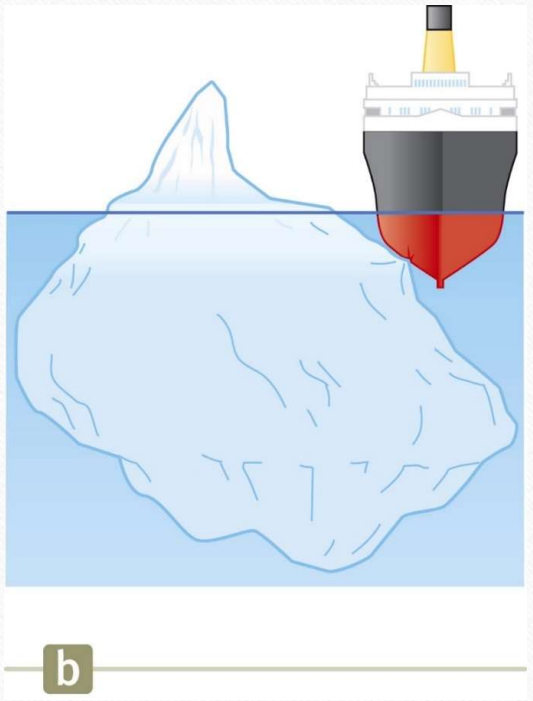
Example 5: Cont.

- $\Sigma F = B + T_2 - F_g = 0$
- $B = F_g - T_2$
- (Weight in air – apparent "weight" in water)
- Archimedes's principle says $B = \rho g V$
  - Find V
- Then to find the material of the crown, $\rho_{\text{crown}} = m_{\text{crown in air}} / V$



Example 6:

- What fraction of the iceberg is below water?
- The iceberg is only partially submerged and so $V_{disp} / V_{ice} = \rho_{ice} / \rho_{seawater}$ applies
- About 89% of the ice is below the water's surface.



Example 7:

- A ping-pong ball has a mass of 2.7g and has a radius of 20 mm. What is the magnitude of its acceleration when placed underneath the surface of a water and released (ρ_{water}=1000kg/m^3)?

  Ignore the effect of water resistance